Evaluation Pipeline for TeamCity's AI Build Analyzer

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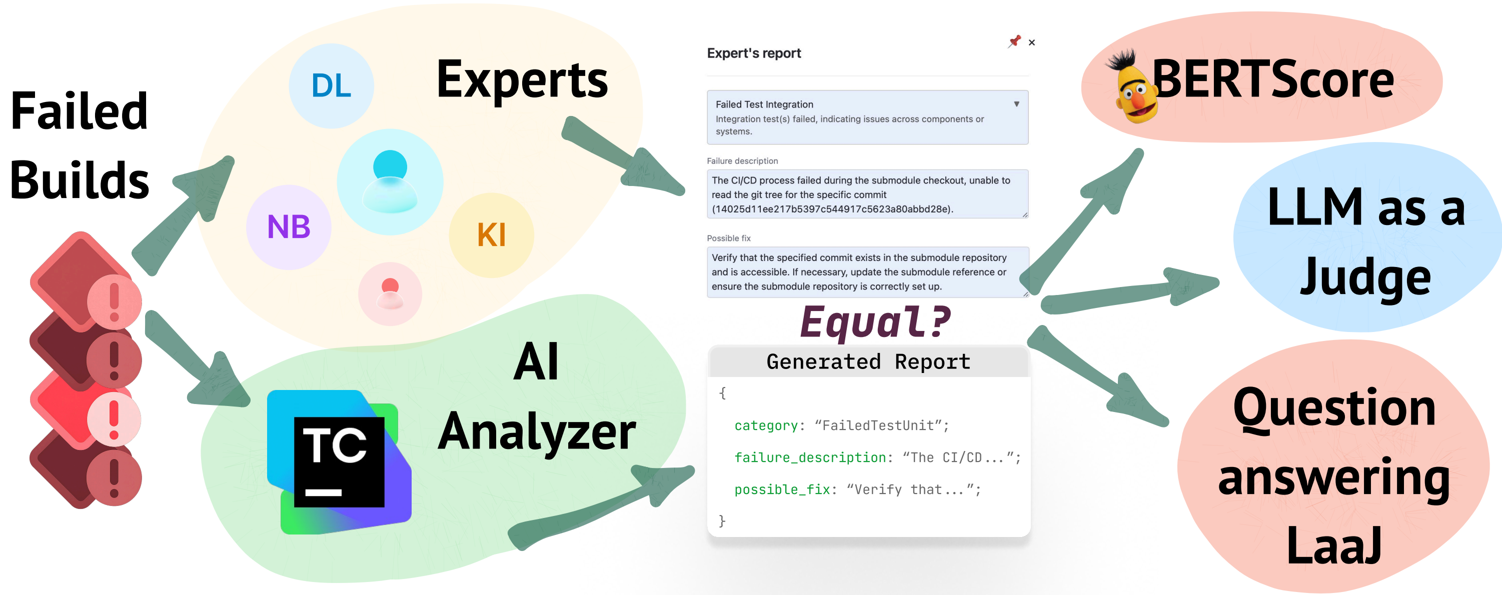
Motivation

- Hard to measure if the agent is truly helpful.
- Reliable evaluation requires real labeled builds.
- No existing tools for appropriate dataset collection and annotation.

Dataset Collection

- We did research on common build failure categories.
- We created a TeamCity plugin to collect annotated failed builds.
- We gathered experts by call-for-action in Slack and Space blogpost.



Metrics Choice

- Best working metrics so far:
  - LaaJ & Question-answering LaaJ
- We ***evaluated*** evaluation metrics by hand-labeling 60 generated reports and computing correlation.
- To expand the dataset, we ***augmented*** that dataset by custom heuristics.
- Achieved ***statistical reliability*** for the eval metrics.

Research Questions

1. How to compare generated reports to ground-truth? Specifically, how to reliably compare two technical texts?
2. Generated report could be more complete than the ground-truth, how to treat that additional info?
3. How to estimate the quality of the metric?
4. How to assess the statistical significance of the result?

Environment setup

- Given two agent versions, we were able to come-up with a setup that guarantees the agent's improvement (*min 5%*) with a defined confidence interval (*by default 95%*).
- We were solving the problem of storing build data and properly interacting retrieving build-information.
- We were integrating the final pipeline into TeamCity CI.

On TeamCity CI

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